

Day 5 –

Q1: Number Guessing Game

```
secret_no = 20

while 'true':

    guess = int(input("Enter your guess :"))

    if guess == secret_no:

        print("Congratulations.")

        break

    else:

        print("Try Again!")
```



```
Enter your guess :10
Try Again!
Enter your guess :23
Try Again!
Enter your guess :20
Congratulations.

Process finished with exit code 0
```

Q2: Skip Multiples of 3

```
for i in range(1, 30):

    if i % 3 == 0:

        continue

    print(i)
```

```
1
2
4
5
7
8
10
11
13
14
16
17
19
20
22
23
25
26
28
29
```

Process finished with exit code 0

Q3: Smart Number Search

```
search_no = 18
```

```
for i in range(1, 50):
```

```
    if i == search_no:
```

```
        print("Number found.")
```

```
        break
```

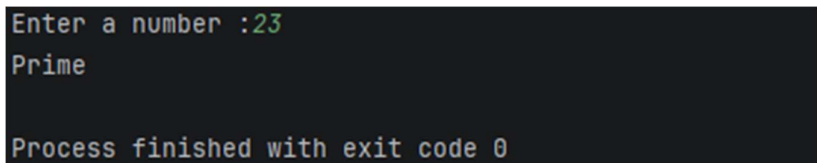
```
Number found.
```

Process finished with exit code 0

Q 4: Prime Number Checker

```
n = int(input("Enter a number :"))
```

```
if n <= 0:
    print("Not prime.")
else:
    for i in range (2, n):
        if n % i == 0:
            print("Not Prime")
            break
    else:
        print("Prime")
```



A terminal window with a dark background. The first line shows the prompt 'Enter a number :23' in green text, followed by the user input '23'. The second line shows the output 'Prime' in green text. The third line shows the message 'Process finished with exit code 0' in green text.

Question 5: Count Vowels in a String

```
Text = input("Enter a String :")
```

```
a=e=i=o=u=0
```

```
for ch in Text.lower():
```

```
    if ch == 'a':
```

```
        a += 1
```

```
    elif ch == 'e':
```

```
        e += 1
```

```
    elif ch == 'i':
```

```
        i += 1
    elif ch == 'o':
        i += 1
    elif ch == 'u':
        u += 1

total = a+e+i+o+u

print("a=",a)
print("e=",e)
print("i=",i)
print("o=",o)
print("u=",u)
print("Total Vowel =",total)
```

```
Enter a String :Pranjal
a= 2
e= 0
i= 0
o= 0
u= 0
Total Vowel = 2

Process finished with exit code 0
```

Q6: Student Name Analyzer

```
name = input("Enter student name :")
```

```
print("Total length:",len(name))  
print("First Character:",name[0])  
print("Last Character :",name[-1])  
print("Reverse Name :",name[::-1])
```

```
Enter student name :Pranjal  
Total length: 7  
First Character: P  
Last Character : l  
Reverse Name : lajnarP  
  
Process finished with exit code 0
```

Question 7: String Index Explorer

```
name = input("Enter a string :")  
print("First Character :",name[0])  
print("Second Character :",name[1])  
print("Last Character :",name[-1])  
print("Middle Character :",name[len(name)//2])
```

```
Enter a string :Pranjal  
First Character : P  
Second Character : r  
Last Character : l  
Middle Character : n  
  
Process finished with exit code 0
```

Q8: Password Strength Checker

```
password = input("Enter your password: ")
```

```
digit = 'false'
```

```
for ch in password:
    if ch.isdigit():
        digit = 'true'
        break

if len(password)>=8 and digit:
    print("Strong Password.")
else:
    print("Weak Password.")
```

```
Enter your password: Pranjal@123
Strong Password.

Process finished with exit code 0
```

Q9: String Utility Program

```
string = input("Enter a String: ")
print("String :",string)
print("Uppercase :",string.upper())
print("Lowercase :",string.lower())
print("Reverse :",string[::-1])
```

```
Enter a String: level
String : level
Uppercase : LEVEL
Lowercase : level
Reverse : level

Process finished with exit code 0
```

Q10: Character Counter

```
text = input("Enter a string :")  
ch = input("Enter a character :")
```

```
count = 0
```

```
for i in text:
```

```
    if i == ch:
```

```
        count += 1
```

```
print("count =",count)
```

```
Enter a string :Sviiiiiii  
Enter a character :i  
count = 8  
  
Process finished with exit code 0
```

Q11: Word Replacement Tool

```
sentence = input("Enter a sentence including 'Java' word :")
```

```
new_sentence = sentence.replace("Java", "Python")
```

```
print("New sentence :",new_sentence)
```

```
Enter a sentence including 'Java' word :I m learning Java  
New sentence : I m learning Python  
  
Process finished with exit code 0
```

Q12: Palindrome Checker

```
word = input("Enter a word :")
```

```
if word == word[::-1]:
```

```
    print("Palindrome.")
```

```
else:
```

```
    print("Not Palindrome.")
```

```
Enter a word :LEVEL
Palindrome.

Process finished with exit code 0
```

Q13:

```
name = input("Enter a name :")
```

```
space = 0
```

```
for ch in name:
```

```
    if ch == " ":
```

```
        space += 1
```

```
        break
```

```
if len(name) > 5 and space == 0:
```

```
    print("Valid Username.")
```

```
else:
```

```
    print("Invalid Username.")
```

```
Enter a name :Pranjal
Invalid Username.

Process finished with exit code 0
```


Question 14: Smart String Dashboard

```
name = input("Enter full name: ")
print("Full Name: ",name)
print("Uppercase: ",name.upper())
print("Lowercase: ",name.lower())
print("Length: ",len(name))
print("First Character: ",name[0])
print("Last Character: ",name[-1])
print("reverse: ",name[::-1])
```



```
Enter full name: Pranjal Ramesh Khade
Full Name:  Pranjal Ramesh Khade
Uppercase:  PRANJAL RAMESH KHADE
Lowercase:  pranjal ramesh khade
Length:  20
First Character:  P
Last Character:  e
reverse:  edahK hsemaR lajnarP
```

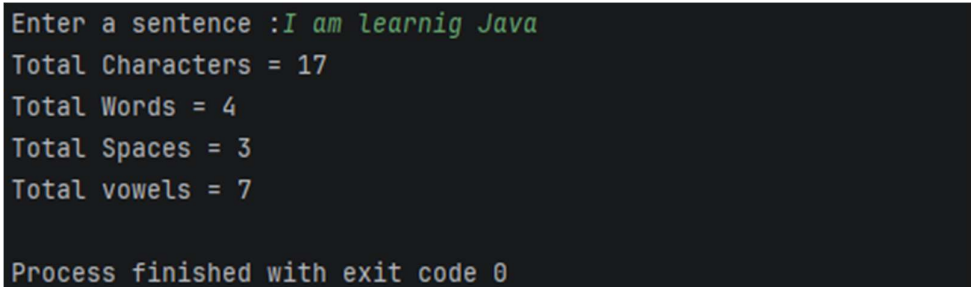
Question 15: Text Analyzer

```
sentence = input("Enter a sentence :")
character = len(sentence)
word = 1
space = 0
vowel = 0
```

```
for ch in sentence:
```

```
if ch == " ":  
    space += 1  
    word += 1  
elif ch == 'a' or ch == 'e' or ch == 'i' or ch == 'o' or ch == 'u' or ch  
== 'A' or ch == 'E' or ch == 'I' or ch == 'O' or ch == 'U':  
    vowel += 1
```

```
print("Total Characters =",character)  
print("Total Words =",word)  
print("Total Spaces =",space)  
print("Total vowels =",vowel)
```



```
Enter a sentence :I am learnig Java  
Total Characters = 17  
Total Words = 4  
Total Spaces = 3  
Total vowels = 7  
  
Process finished with exit code 0
```

Question 16: String Utility Dashboard

```
name = input("Enter name: ")  
  
print("-----Menu-----")  
print("1. Convert to Uppercase.")  
print("2. Convert to Lowercase.")  
print("3. Reverse String.")  
print("4. Find Length.")
```

```
print("5. Count Vowel.")
```

```
print("6. Check Palindrome.")
```

```
choice = int(input("Enter your choice: "))
```

```
if choice == 1:
```

```
    print("Name in Uppercase :",name.upper())
```

```
elif choice == 2:
```

```
    print("Name in Lowercase :",name.lower())
```

```
elif choice == 3:
```

```
    print("Reverse :",name[::-1])
```

```
elif choice == 4:
```

```
    print("Length :",len(name))
```

```
elif choice == 5:
```

```
    vowel = 0
```

```
    for ch in name:
```

```
        if ch == 'a' or ch == 'e' or ch == 'i' or ch == 'o' or ch == 'u':
```

```
            vowel += 1
```

```
    print("Count of Vowels :",vowel)
```

```
elif choice == 6:
```

```
    if name == name[::-1]:
```

```
        print("Palindrome.")
```

```
    else:
```

```
print("Not Palindrome.")
```

else:

```
print("Invalid Choice!")
```

```
Enter name: Ram
-----Menu-----
1. Convert to Uppercase.
2. Convert to Lowercase.
3. Reverse String.
4. Find Length.
5. Count Vowel.
6. Check Palindrome.
Enter your choice: 6
Not Palindrome.

Process finished with exit code 0
```